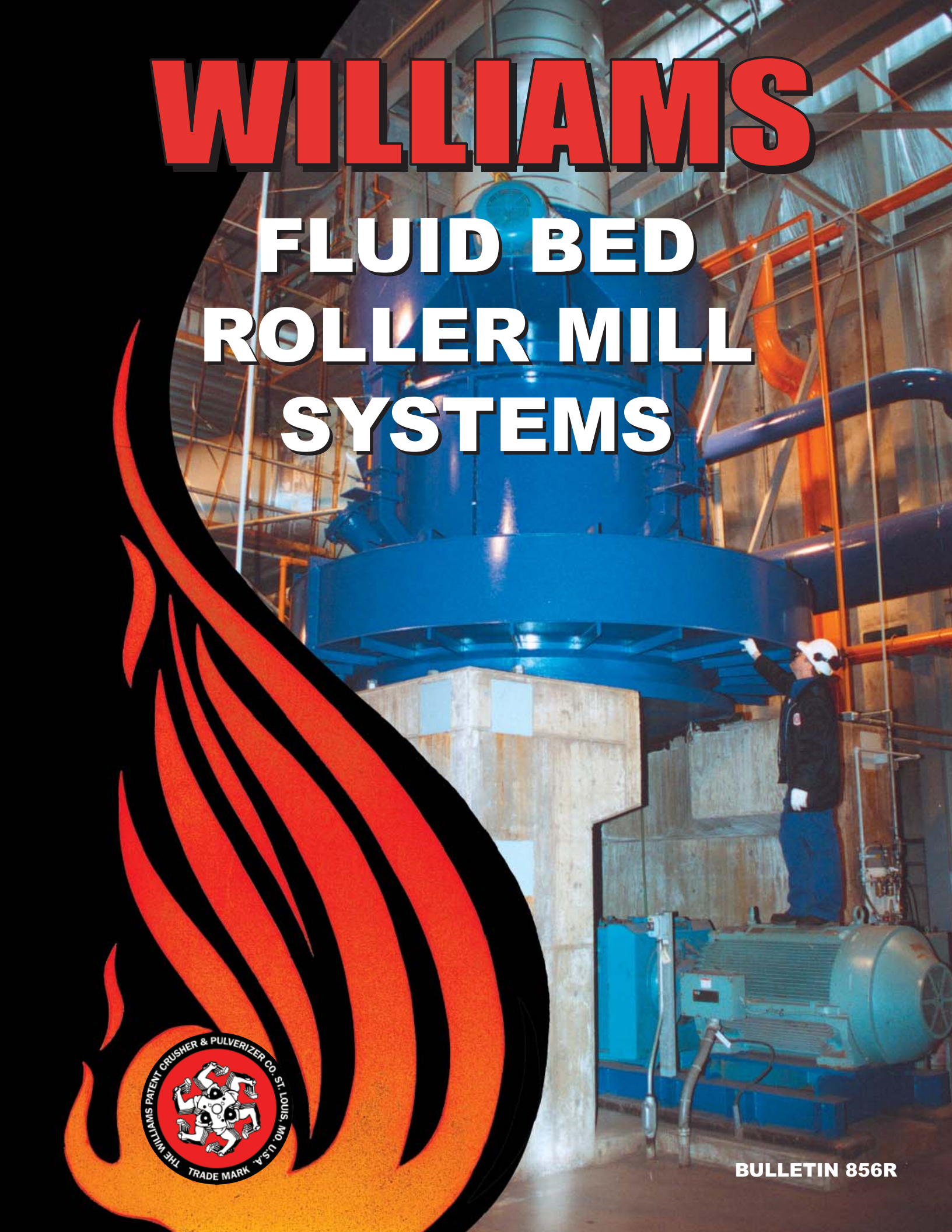
WILLIAMS

FLUID BED ROLLER MILL SYSTEMS



THE WILLIAMS PATENT CRUSHER & PULVERIZER CO. ST. LOUIS, MO. U.S.A. TRADE MARK

BULLETIN 856R

Williams is the acknowledged leader in Fluid Bed Roller Mill Design. Here are a few reasons why...

1. TURN DOWN RATIO
Infinite turn down while maintaining product size. Williams adjusts the mill grinding rate as a direct function of control system demand. The mill capacity is modulated by varying mill speed.

2. EXTERNAL GEAR BOX
External right angle gear box eliminates contamination of product into gear box. Removal of gear box from skid allows change out of mill main vertical shaft and saves money by sizing the transmission to the required work.

3. NO SPRINGS
No springs or external pressure adjustments are required for product size change or to compensate for wear. Centrifugal force is utilized to control the grinding. This force is adjusted through instantaneous response of the mill speed which in turn compensates for grindability and flow.

4. MILL SPEED
Mill speed modulation varies the centrifugal force- product capacity ranges from 100% to 0% utilizing an optional variable frequency AC drive.

5. PRODUCT CONTROL
Micrometer control of finished product. Product size (regardless of turn down or grinding element condition) is controlled by utilizing the Williams Spinner Separator equipped with a variable speed drive. Drive adjustment is controlled automatically by a patented control system.

6. MOTORS
All drive motors are standard TEFC, premium efficiency motors. All drives are directly connected through standard flexible coupling, or if desired, by V-belt drive.

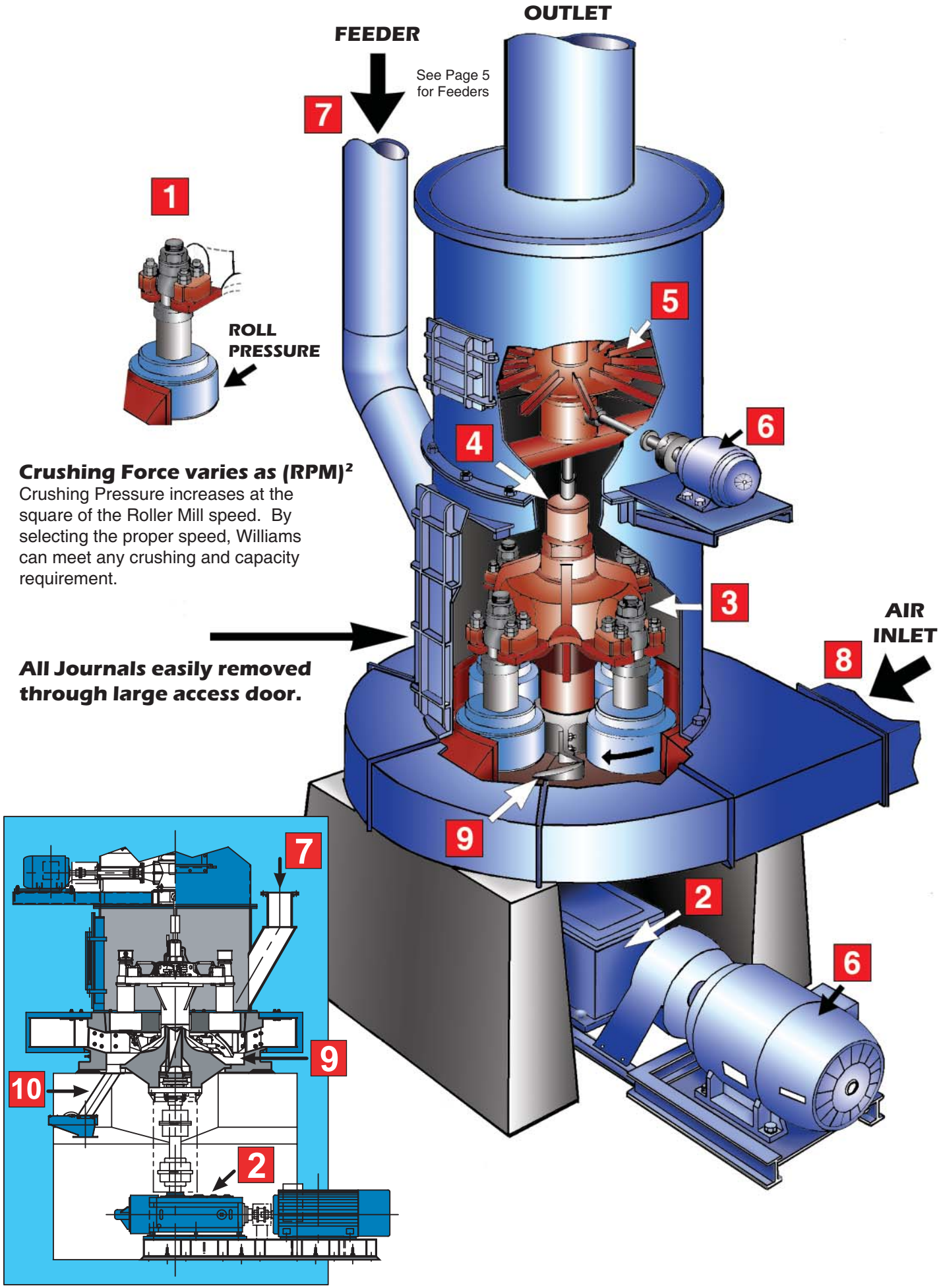
7. FEEDER
A wide variety of feeders is available to best suit the material being processed. These are Weigh Belt Feeders, Live Bottom Bin Feeders, Screw Feeders, Vibratory Feeders, or Rotary Feeders.

8. FLUIDIZED BED DRYING
Hot gasses entering the grinding chamber below the grinding rolls fluidize the entire grinding chamber and promote rapid and efficient drying of all material types, including lignite, clays, and sorbent. Large open area through grinding chamber allows particle termination velocities to be maintained while fluidized by heated gasses.

9. PLOW CLEARANCE
Adequate clearance between the plows and the mill base provides a “stone box” effect eliminating the need for liners in the area. This clearance is an added margin for protection against tramp metal.

10. BOTTOM DISCHARGE
Auxiliary Mill Bottom Discharge permits production of a second granular product or beneficiation of the principle product- either can be done simultaneously with fine grinding and drying.





Crushing Force varies as (RPM)²
Crushing Pressure increases at the square of the Roller Mill speed. By selecting the proper speed, Williams can meet any crushing and capacity requirement.

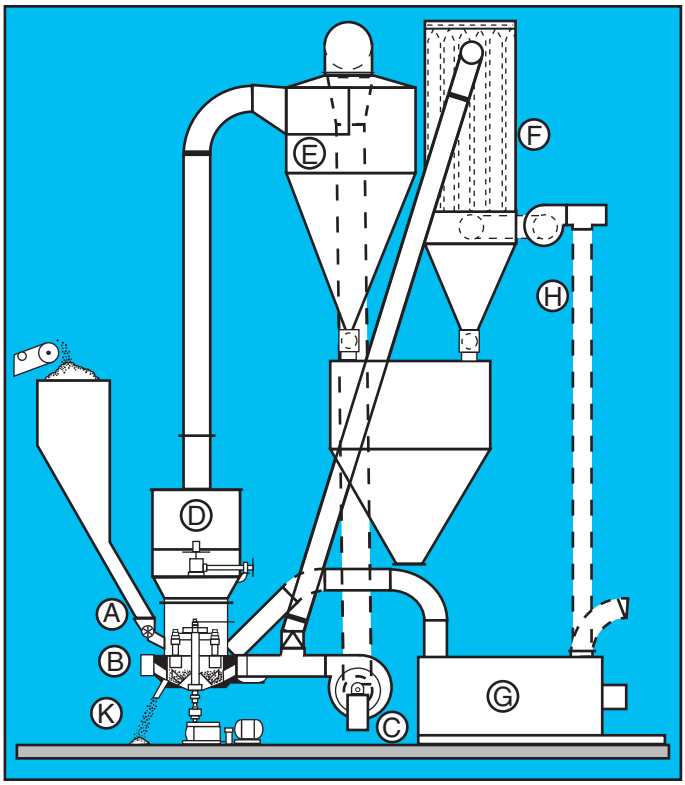
All Journals easily removed through large access door.



OPERATION OF THE ROLLER MILL SYSTEM

Operation of the Roll Mill System is completely automatic. Feeder (A) introduces raw material into the grinding chamber in the base of the Roller Mill (B) at a rate determined by pressure variations within the Roller Mill.

Plows located ahead of each roller direct the material upward and between the grinding rolls and the heavy alloy steel bull ring where it is ground to size. The grinding rolls are free to swing out centrifugally, which forces them to bear upon and grind the raw material. Being free to pivot, the rolls automatically assume the proper position for grinding and need no adjustment to compensate for wear.



Once ground to size, material is swept out of the mill by the controlled air flow from the centrifugal main mill fan (C). Air suspended, the ground material passes through the adjustable Williams Separator (D). In the separator, an accurate size classification is made which allows product sized material to be conveyed away from the Roller Mill while oversized material is returned for additional grinding.

Product sized material is removed from the airstream by the Cyclone Collector (E). Typically, a conveyor or product storage bin will be located to accept the product as it exits from the bottom of the Cyclone Collector.

Airborne fines which remain entrained in the exhaust gas from the Cyclone Collector are removed by a high efficiency fabric dust collector located downstream from the cyclone. The efficiency of the Dust Collector (F) assures conformance with environmental codes.

The basic Roller Mill System becomes a highly efficient means of removing moisture from the raw material by the simple addition of an air heater (G).

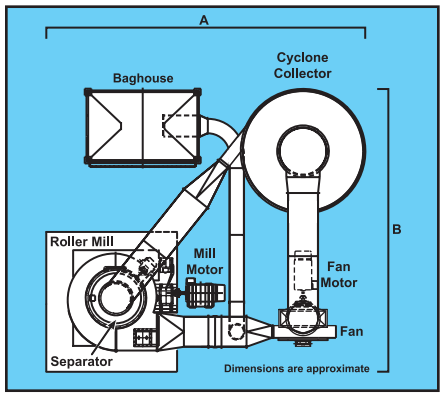
Further refinements to the heater system permit the Roller Mill to process hazardous dusts in a controlled low oxygen atmosphere. A recycling line (H) from the dust collector to the heater enables a further reduction in oxygen level. This is Williams' patented Inert Gas System.

Product beneficiation is possible in the Williams Roller Mill System by continuous removal of impurities through a discharge spout (K) located beneath the Roller Mill.

DIMENSIONS & ARRANGEMENTS

The various parts of a Williams Roller Mill, with its accessory air equipment, can usually be placed to fit into most industrial type buildings. The purchaser supplies the Williams Engineering Department with a sketch of the building. The system components are then engineered to fit into the available space.

This drawing and the table show the minimum space required for each size unit. The Cyclone Collector and Auxiliary Collector can be placed higher enabling the discharge to dump into a storage bin.

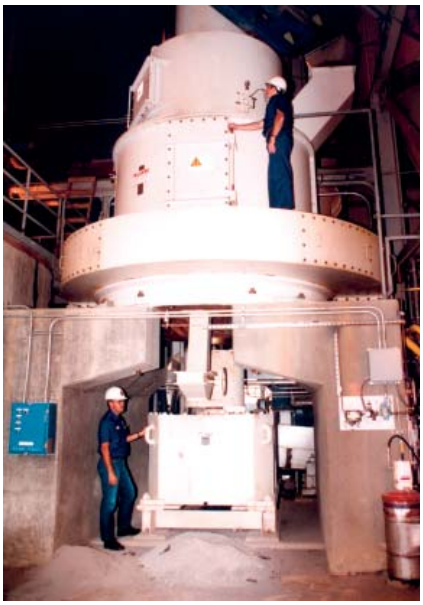


ROLLER MILL SIZING CHART

Roller Mill Size	A	B	Height	Capacity Factor	Mill	Fan	Separator	Feeder	Approx. Weight (lbs)
Laboratory	12'-0"	9'-0"	18'-0"	.5%	3	15	1	1/3	3,900
Zenith	17'-0"	10'-6"	20'-6"	1.7%	7.5	15	1	1/3	6,900
Gnome	18'-6"	11'-6"	22'-9"	5.5%	25	20	1	1/2	12,600
Comet	19'-8"	14'-6"	28'-0"	9.0%	30-40	25-30	3	1	15,000
Standard	24'-0"	21'-0"	32'-6"	24%	60-75	60-75	5	1	22,000
Super-Titan	32'-0"	25'-0"	37'-0"	40%	100-150	100-150	5-7 1/2	2-3	47,000
Champion	34'-0"	33'-0"	40'-0"	57%	200-250	150-200	7 1/2	5	90,000
Giant	38'-0"	33'-0"	45'-0"	68%	250-300	150-200	10-15	5-7 1/2	105,000
Super-Giant	40'-0"	37'-0"	48'-6"	100%	350-450	200-250	15-20	7-1/2	110,000
Hercules	42'-6"	37'-0"	51'-0"	123%	350-500	300-400	20	10	120,000
Colossus	45'-0"	39'-6"	53'-0"	144%	400-600	350-450	25	10	136,000
Mammoth	41'-0"	40'-0"	56'-0"	168%	500-700	400-500	25	10	153,000
Nova	51'-6"	47'-0"	60'-6"	222%	600-800	500-600	30	15	192,000
Super Nova	56'-0"	52'-0"	65'-0"	286%	800-1000	700-800	30	13	236,000
Saturn	58'-0"	56'-0"	70'-0"	357%	1000-1500	1000-1200	50	20	375,000
Jupiter	60'-0"	60'-0"	75'-0"	467%	1250-2000	1000-1500	50	25	475,000
Zeus	70'-0"	70'-0"	80'-0"	591%	1500-2500	1250-1500	75	30	585,000
Super Zeus	75'-0"	75'-0"	85'-0"	730%	2000-5000	1500-2500	100	50	685,000

PERFORMANCE DATA

Material	Fineness at Rated Capacity (M=Mesh)	Capacity Tons per Hour (Super Giant)	Feed Moisture (%)	Product Moisture (%)
Barytes	92%-325M	35	5-8	-1.0
Bauxite	99%-325M	13	19	1.0
Bentonite	85%-200M	26	15	7
Clay				
Fullers Earth	99%-3/4"	50-60	47	35
Calcined	97%-325M	15	0	0
Ceramic	97%-30 Micron	8-9	25	1.0
Coal				
Calcined Anthracite	99%-40M	31	0	0
Western Bituminous	80%-200M	20-25	18	5
Eastern Bituminous	70%-200M	30-35	15	1.0
Lignite	80%-200M	44	19	4
Coke				
Shot	90%-200M	25	5	0
Petroleum	95%-200M	35	5	0
Green Delay	95%-200M	45	9	1.0
Dolomite	85%-200M	30	0	0
Dross, Aluminum Zinc	99%-100M	18-20	0	0
Oxide, & Zinc Powder	30%-35M	18-20	0	0
Ferro Silicon	99.7%-325M	15	0	0
Graphite	80%-325M	10	0	0
Gypsum	99%-100M	70	10	0
Iron Oxide	100%-325M	10	0	0
Kaolin	99.99%-325M	10-20	0	0
Lime, Burnt	99%-200M	10-20	0	0
Limestone				
Medium Hard	95%-200M	30-36	8	0
Coral	90%-100M	28	8	0
"Mine Dust"	70%-100M	35	8	0
Magnetite Ore	92%-325M	8-10	5	0
Magnetite Ore	70%-325M	12-15	5	0
Perlite, Expanded	50%-325M	25	0	0
Phosphate Rock	50%-200M	35-38	5-8	0
Phosphate Rock	70%-200M	30-32	5-8	0
Shale	98%-70M	40	4	0
Silica, Fused	70%-325M	20	0	0
Sponge Iron	70%-100M	18-20	0	0
Sulphur	95%-325M	25-28	0	0
Talc	98%-325M	30	0	0
Titanium Oxide	99.95%-325M	5-7	0	0
Tobacco	99%-100M	5-6	0	0





* Capacities shown apply to the Super Giant size Roller Mill. Use Capacity Percentage Factor for smaller or larger Mills.

COMPONENTS FOR COMPLETE ENGINEERED SYSTEMS



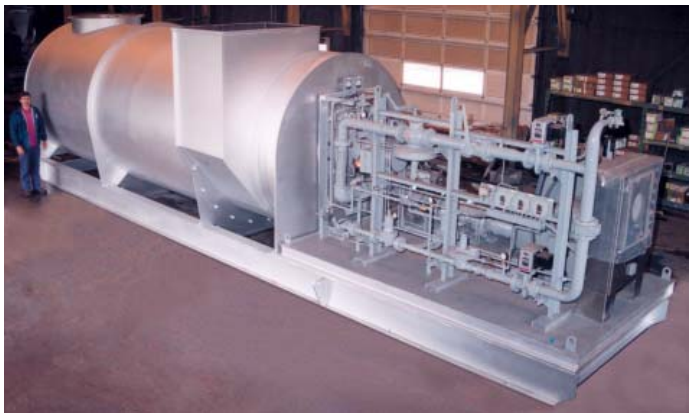
Large Air Systems are available up to 400,000 ACFM including Primary and Secondary Fans, Cyclones, and necessary pipework, elbows, and transitions to connect and integrate into the system.



Heavy duty, vacuum type **Cyclone Collectors** for efficient removal of sized product from the airstream.



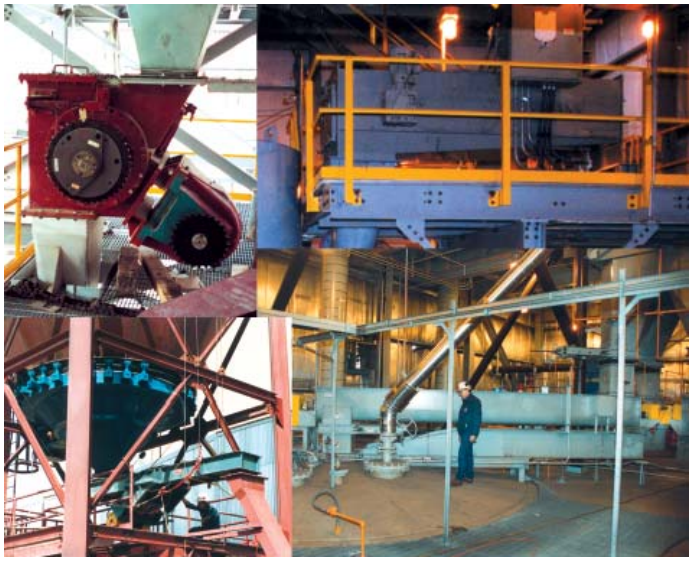
Splitter Valves Roller Mill product can be directed into numerous streams by utilizing splitter assemblies.



Williams Air Heaters, when added to the Roller Mill System, permit the operator to dry material simultaneously with the grinding operation. Heaters are designed to comply with insurance company regulations and are available for firing with gas, oil, or dual fuel. With modification, the heater becomes a part of Williams' patented Inert Gas System for grinding hazardous materials such as coal.



Dust Collectors Baghouses and Precipitators can be provided for exhaust systems to meet the most stringent requirements.



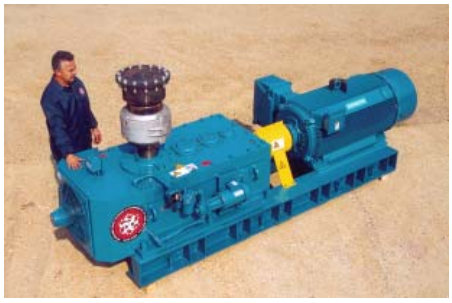
Feeders Williams Feeders are designed to work in conjunction with our other systems and for long service life under severe loads.

BENEFITS OF WILLIAMS ROLLER MILLS

Low Maintenance & Easy Service Access



Large Access Doors for inspection and maintenance. Journals are easily removable through the access door so maintenance can be done outside the mill.



Remote, External Drive Assembly allows proper selection of the transmission to the actual work requirements. Spacer coupling insures proper alignment and prevents excess heat transfer or product contamination from effecting the gearbox.

Four Types of Air Classifiers

Williams offers four different types of Air Classifiers to compliment our Roller Mills:

Spinner Separators are simple air classifiers incorporated into an existing air system or an air swept mill. This allows precise control of the top size as it leaves the air column.

High Efficiency Turbine Separators are also used with air swept mills where efficiency and ultra fineness are a consideration.

Isokinetic Separators are used when trying to produce a product that is very steep on the product curve, such as fluid bed sorbent products.

Velocity Separators are the simplest in design and are utilized in processes where sizing does not have to be so exact.





Williams Complete Control Systems



Williams has the capability of providing complete system controls utilizing the latest generation of hardware and software which offer up-to-date economical solutions for your process optimization.

The Williams S7-2000 is a highly stable controller based on a Siemens S7 platform using Williams' custom designed PID loops. The system is provided pre-programmed and customized for your specific application. All of our control systems can be loaded with program modifications and upgraded easily through our sophisticated HMI software package, also standard.

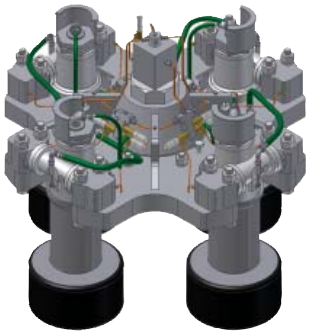
The Williams S7-2000 can improve process monitoring and data acquisitions, at a very affordable price. It can also generate network access process information with minimum disruption to production or current operations. System communication interfaces available include Profi-bus, Modbus, OPC Server, Ethernet, Modbus over Ethernet, and Profi-bus over Ethernet.

A Microsoft Windows™ based HMI system provides a familiar, and user friendly environment for facilitating quick and easy training. The system comes designed with a touch screen interface, allowing the user to look at multiple screens and Maintenance Manuals including Parts Lists, which are stored in the tech screen software. Our standard unit comes with blocks of 48 input and output discretes, 16 analog inputs and outputs, and 8 thermo coupled inputs. Additional blocks can be added as required. Williams patented control software (U.S. Patent #6,467,707) allows the user infinite turn down and multiple products from the same system.

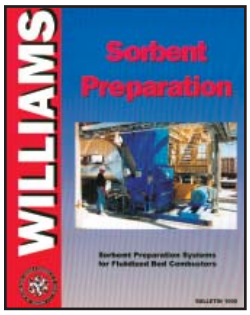
Continuous Automatic Lubrication System

All Williams Roller Mills incorporate an automatic lubrication system, which provides continuous lubrication to all internal parts of the machine while in service. This eliminates the need to shut the machine down for manual lubrication. The auto lubrication system can be controlled by our S7-2000 controller or an independent stand alone PLC. Our latest design incorporates a Lincoln manifold type injector, which allows the lubricant output to design to be varied from .008 in^3 to .08 in^3 per cycle.

Our automated controller is fully programmable, which allows complete customized cycle times based on your specific application. The automated controller also monitors each cycle and will alert the operators in the event a fault is detected.



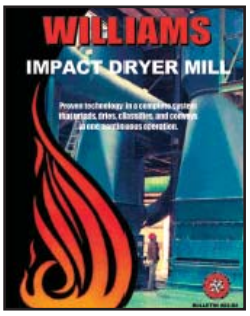
Call us to see our complete line of Williams Crushers & Equipment!



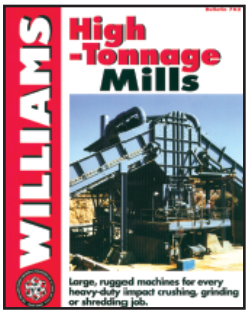
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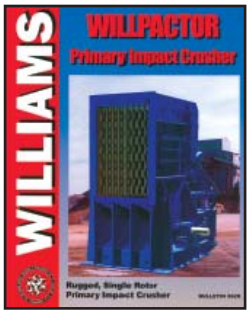
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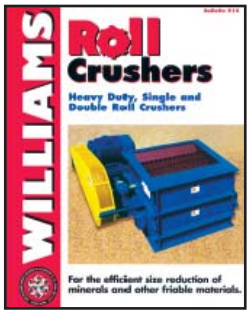
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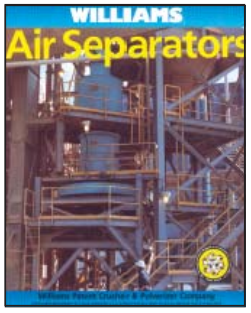
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BULLETIN 915



BULLETIN 998R



BULLETIN 774

The Williams Way...

Williams Patent Crusher is a 125-year-old family owned business that custom designs and manufactures size reduction equipment and related process systems for worldwide industrial markets. We strive to achieve industrial leadership through excellence in product quality, customer service, innovation, employee involvement, and measured growth. Our staff of highly qualified engineers combines the benefits of irreplaceable practical experience with the sophisticated accuracy of our own CAD/CAM interactive graphics computer system with finite analysis, which permits three-dimensional, "on-screen" design of parts and translates them into numerical language for actual machining to keep us in the forefront of modern manufacturing. We promote a team environment that will allow us to recognize and respond to an ever-changing marketplace. See your Williams representative or contact us directly.

Some of the materials being processed in Williams Roller Mills:

- Aluminum Dross
- Barytes
- Bauxite
- Bentonite
- Burnt Lime
- Calcium Carbide
- Carbon Activated
- Chalk
- Chemicals
- Clay
- Coal
- Copper Oxide
- Diatomaceous Earth
- Dolomite
- Ferro Silicon
- Feldspar
- Fullers Earth
- Graphite
- Gypsum
- Iron Ore
- Iron Oxide
- Kaolin
- Licorice
- Limestone
- Lithium Ore
- Marble
- Nickel/Aluminum Alloy
- Oyster Shells
- Perlite
- Petroleum Coke
- Phosphate Rock
- Phosphate Rock Clay
- Pigments
- Pumice
- Resin
- Sulphur
- Talc
- Tire Char
- Titanium Dioxide
- Tobacco
- Zinc Dross

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